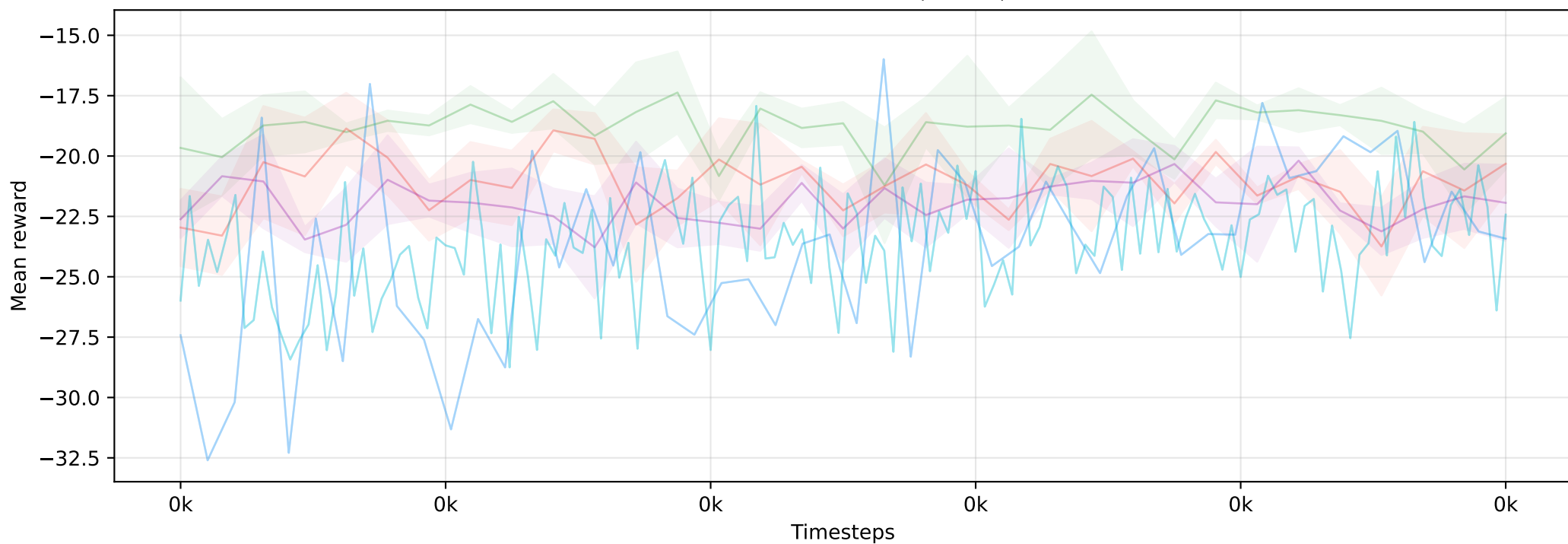
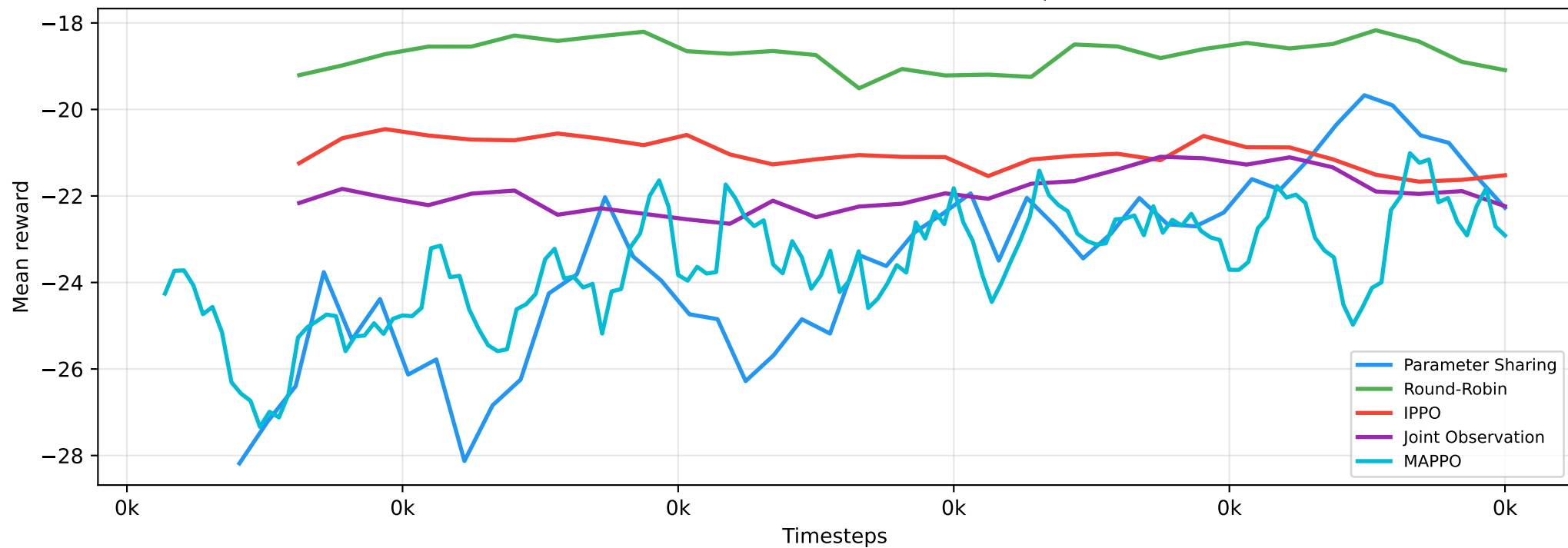


PPO Evaluation — Mean Episode Reward

Raw eval mean reward (± 1 std)

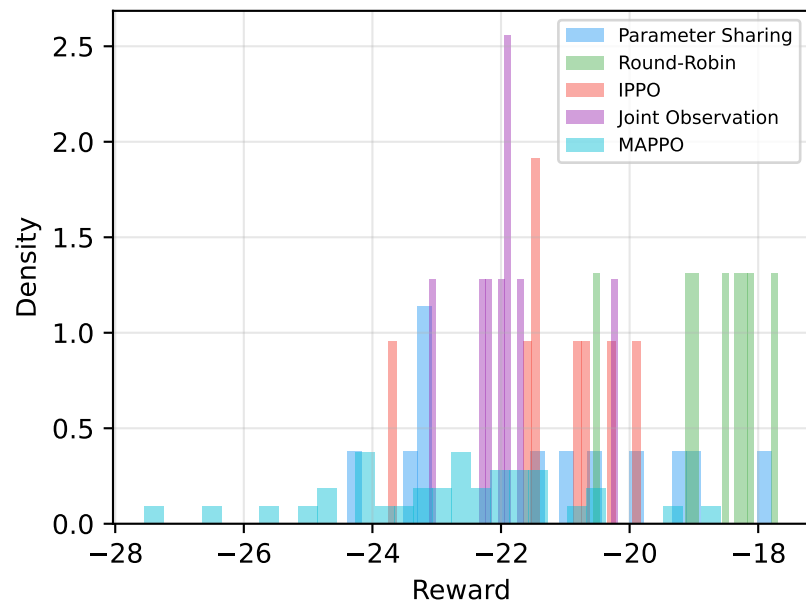


Smoothed eval mean reward (window=5 checkpoints)

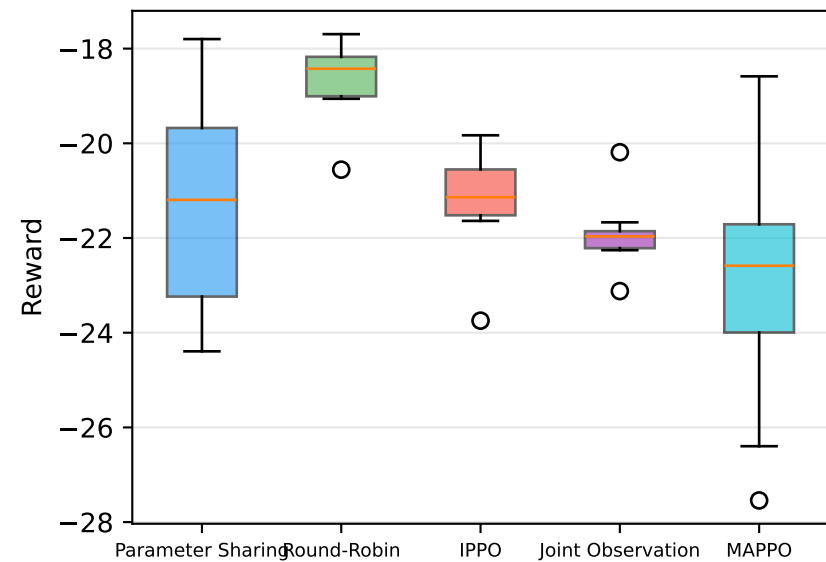


PPO Evaluation — Distribution & Convergence

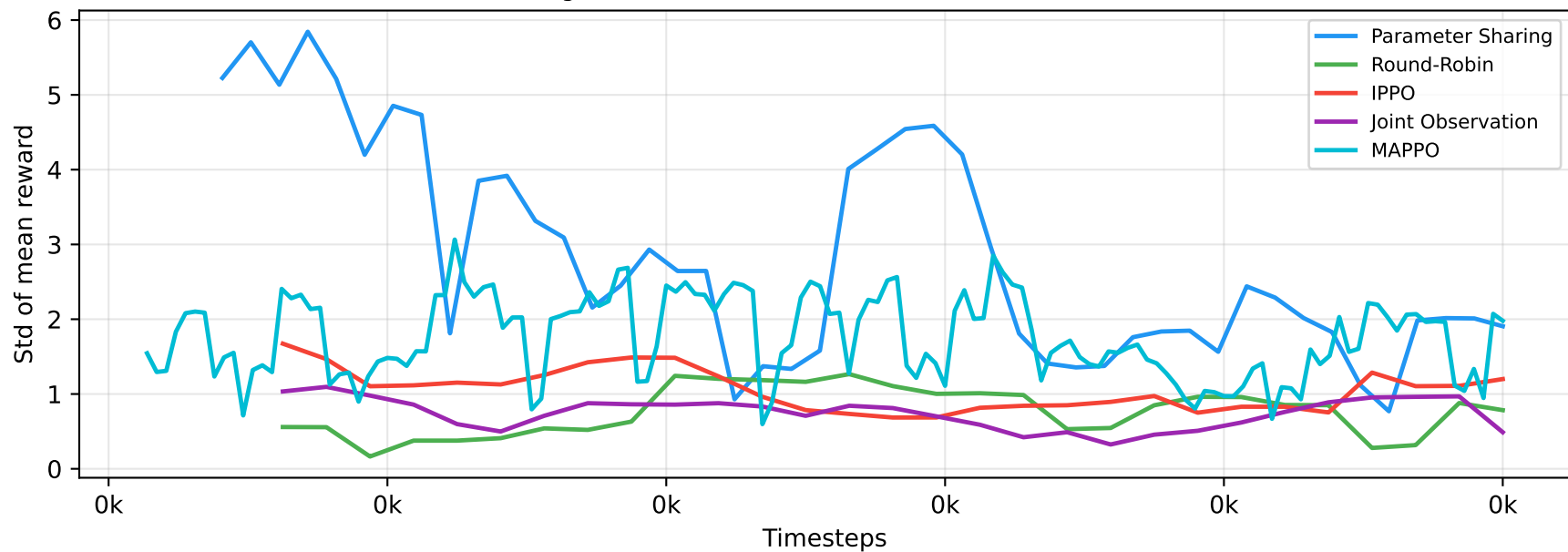
Reward distribution
(last 25% of evals)



Reward spread
(last 25% of evals)



Rolling reward std (window=5) — lower = more stable



PPO Evaluation — Summary Statistics (last 25% of evals)

Run	Total steps	Eval checkpoints	Mean reward	Std	Max	Min
Parameter Sharing	0k	50	-21.351	2.040	-17.799	-24.393
Round-Robin	0k	33	-18.681	0.826	-17.695	-20.556
IPPO	0k	33	-21.240	1.112	-19.831	-23.746
Joint Observation	0k	33	-21.911	0.766	-20.189	-23.121
MAPPO	0k	146	-22.790	1.831	-18.586	-27.542